

# FlavorMap

## แผนที่รสชาติ

### The Project Bible — v3, Implementation Edition

Computational Geography of Thai Cuisine · Rescoped against a real calendar and a real hour budget  
ISB Bangkok · Grade 11 → Grade 12 · Pillar 2 of the Future Leader Plan · **Supersedes v2**

**Why this version exists.** v2 fixed the authorship problem and was right to. But it was written against a 12-month clock that no longer applies, it targeted a conference cycle that has passed, and — most importantly — it sized the fieldwork for a job the fieldwork no longer has to do. v3 changes no methods. It changes scope, sequence, and what depends on what.

The single structural correction: **no research question may depend on fieldwork completing.** v2's risk register named cancelled fieldwork as the failure mode that "destroys the entire thesis." A project whose thesis can be destroyed by one cancelled weekend is badly built. In v3 the fieldwork corroborates rather than carries — and is therefore smaller, deeper, and far more likely to actually happen.

**Three items block everything else.** They are listed in §21 and none of them takes more than an afternoon. Until the labelled-recipe fraction is measured, the shape of this paper is unknown and every plan in this document is provisional.

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## 1. What Changed From v2, and Why

| Area                 | v2                                                                       | v3                                                                               | Reason                                                                                                             |
|----------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Research questions   | Five, phrased around methods (Louvain, Mantel, Moran's I, Random Forest) | Five, phrased around findings; methods demoted to instruments                    | A question that asks whether an algorithm agrees with an administrative map produces a fact about Thai bureaucracy |
| Fieldwork dependency | Two of five RQs required it                                              | Zero RQs require it; it validates RQ3 and enriches RQ2/RQ5                       | Most likely silent failure mode should not be load-bearing                                                         |
| Interview count      | 15–25 across 4–6 provinces                                               | <b>12–15 across 2 provinces</b> (Nan + Surin)                                    | 4 per province supports no claim. 6 per province supports an existence claim                                       |
| Cook-alongs          | 15 dishes                                                                | <b>8 dishes</b>                                                                  | Buys back ~14 hrs for lexicon work, the real bottleneck                                                            |
| Build log            | 10 posts                                                                 | <b>6 posts</b> , two of them failure posts                                       | Log stall was already flagged as a common failure; GitHub history carries most of the authorship evidence          |
| Venue                | DH 2026 student track                                                    | <b>CHR 2028 + Journal of Cultural Analytics</b>                                  | DH 2026 has passed. Nov-2027 applications leave room for a real journal cycle                                      |
| Journal submission   | "Do not attempt – wrong instrument for the timeline"                     | <b>Attempt it.</b> "Under review" is a stronger honours line than a poster       | v2 assumed a 12-month clock. The actual runway is ~15 months to freeze + 8 months to applications                  |
| Infrastructure       | Colab free tier, flat files                                              | Local PostgreSQL 15 + PostGIS in Docker; API serves pre-materialised static JSON | Storage cap and latency on the original plan; static JSON removes the live-query dependency entirely               |
| Dataset release      | One bundle, OpenFlavorTH                                                 | <b>Two HuggingFace repos</b> – corpus and lexicon, released separately           | The lexicon will out-download the corpus and deserves independent citation                                         |

| Area             | v2                                                                         | v3                                                                    | Reason                                                                                               |
|------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Second annotator | LOW-MED priority, "where possible"                                         | <b>Required.</b> 100 double-labelled recipes, report Cohen's $\kappa$ | "Single annotator" is the first objection a reviewer raises; $\kappa$ disposes of it in one sentence |
| Pre-registration | Absent                                                                     | <code>docs/hypotheses.md</code> , dated, committed before analysis    | Makes the negative-results section credible rather than retrofitted                                  |
| Methods stack    | NetworkX, Louvain, TF-IDF, UMAP, K-means, Moran's I, Mantel, Random Forest | <b>Identical.</b> Three statistical corrections applied (§4)          | Nothing removed. Never was the problem                                                               |

The contradiction v3 resolves. v2 exists because v1 had no visible teenager in it, and the fix was fieldwork — irreplaceable, unfakeable, hers. Then feasibility pressure pushed the research questions toward web-only analyses, which is correct for the calendar and quietly undoes v2's whole purpose: five questions a competent adult could answer with a laptop.

v3 does not resolve this by reverting. It gives fieldwork a role that is small enough to survive a cancelled trip and central enough to matter: **ground-truthing the coverage gap**. RQ3 produces a map of Thailand with a third of it blank. Twelve interviews demonstrate what is inside the blank. That is an existence claim, which  $n=12$  fully supports, and no reviewer can object to it because it is not being asked to generalise. See §8.

## 2. The Fingerprints Problem — Retained From v2 UNCHANGED

The reasoning in v2 §1 stands in full and is not restated here. The summary: admissions readers discount sophisticated teen projects steeply because they cannot verify authorship, so **believability is the scarce resource, not sophistication**. The three signals a reader uses are process visibility, irreplaceable input, and calibrated claims. v3 preserves all three mechanisms — public repo from day one, unpolished early notebooks left intact, a named credit ledger, a limitations register written before anyone asks, and fieldwork only she can do.

One correction to v2's framing. v2 said making the project "look more like a high schooler's work" makes it stronger. That is right about evidence and wrong if applied to scope. v3 cuts scope in several places — but every cut is made for hour-budget reasons and is stated as such, never to appear more modest. Under-claiming is a credibility purchase; under-building is not.

## 3. The Origin Story — Still Has an Open Bracket BLOCKING

v2 §2 specified that every description of the project opens with cooking and arrives at code, and supplied a canonical origin paragraph with a bracket in it: the real dish, the real family version.

**That bracket is still open.** Until it is filled with something that actually happened, the essay, the paper's introduction, the first build-log post, and every interview answer rest on a placeholder. This is the one item in the entire document that cannot be reconstructed later — a memory cooked and written down in month 1 is evidence; the same memory reconstructed in month 14 is a story.

**Action:** cook the dish with the family member this month. Photograph the mise en place. Write down what actually happens, including the parts that contradict the remembered version — especially those. Commit the notes and photos to the repo with the real date. Two hours. See §21.

The voice rules from v2 are retained without change: cooking first and method no earlier than the third sentence; first person and concrete verbs; name the false start; no grand framing verbs; keep the sister visible where she contributed.

4. Technical Non-Negotiables + Three Corrections REVISED

The methods stack is unchanged from v1 and v2 — NetworkX, python-louvain, TF-IDF, UMAP, K-means, PySAL/esda, Mantel, Random Forest, PyThaiNLP. Nothing is removed and nothing is simplified. A self-taught 16-year-old correctly applying real tools is the profile that wins; a 16-year-old using simplified tools is just a 16-year-old.

Three corrections apply to how those tools are used. Each is a mistake that would have survived review at a student track and been caught at a serious venue.

| Correction          | What v1/v2 implied                                                                                                   | What v3 requires                                                                                                                                                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mantel significance | A Mantel test evaluated with a parametric p-value (e.g. <code>scipy.stats.pearsonr</code> on the flattened matrices) | <b>Permutation testing, 9,999 permutations.</b> Distance-matrix entries are not independent observations; a parametric p-value on them is meaningless and a reviewer will say so immediately                                                                            |
| Moran's I inputs    | "Moran's I on province-level flavour scores" — ambiguous enough to invite running it on UMAP dimensions              | <b>Only on interpretable scalars</b> — prevalence of a named ingredient, distinctiveness score, recipe count. Never on a UMAP dimension, which has no units and no meaning                                                                                              |
| UMAP interpretation | "What does the recipe space look like in two dimensions?"                                                            | <b>UMAP axes carry no interpretable meaning.</b> Label them "UMAP 1 / UMAP 2", caption them as uninterpretable, never quantify from the projection, never call an axis geographic or cultural. Use it as a browsing interface on the site, not as a result in the paper |

**On dimensionality.** Every chart in this project is 2D. Three-dimensional scatter plots lose the ability to read any point's position, introduce occlusion, and are unreproducible in a paper because the figure depends on a chosen camera angle. Where a third spatial axis is tempting, use colour, size, or small multiples. **The single exception is the force-directed ingredient network on the website,** where 3D is not clearer but is worth spinning. The paper's network figure stays 2D.

## WHAT GETS ADDED, NOT SUBSTITUTED

- **A negative-results section** at full weight, not in a footnote. Cheapest credibility available.
- **A pre-registration.** [docs/hypotheses.md](#) , committed and dated before any analysis runs, stating what she expects for each RQ. Then report where she was wrong. Trivially cheap, unusual for a student project, and it is what makes the negative results read as findings rather than as excuses.
- **A bias audit of the corpus itself**, quantified rather than mentioned — now promoted to a full research question (RQ3).
- **Inter-annotator agreement.** 100 recipes independently labelled by a second Thai reader; report Cohen's  $\kappa$ .

## 5. The Five Research Questions REWRITTEN

Each question is stated twice: the precise form for the paper, and the plain form for the site, the build log, and interviews. These are not two audiences being compromised between — she needs both phrasings anyway, and writing them together prevents the plain version from drifting past what the data supports.

Each also carries the **transferable claim**: what a reader outside Thai food studies takes away. A question that produces only a fact about Thai cuisine gets a poster. A question that produces a method or a general claim gets a talk.

### RQ1 — Are cultural boundaries discrete or continuous, and can they be located from compositional data alone?

Plain: how far do you travel before the food changes — and does it change gradually, or all at once?

|                 |                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method          | Distance-decay curve (cosine distance on province TF-IDF vs. great-circle km) with change-point detection. Louvain communities compared against <b>one</b> competing boundary set. |
| Output          | A number in kilometres: the width of the boundary zone. Figure 2.                                                                                                                  |
| A NO looks like | Smooth decay everywhere, no discontinuities — meaning the four-region model is an administrative fiction over a continuum. More provocative than the positive result, not less.    |
| Transferable    | A method for locating cultural boundaries without assuming them, applicable to any spatially-indexed compositional corpus: dialect atlases, folk music, craft traditions.          |

**Scope note.** v2 implied testing administrative and historical and linguistic boundary sets. Digitising pre-1900 polity boundaries is a small project on its own. **Pick one: linguistic.** Ethnolinguistic maps of Thailand already exist, and the upper-vs-lower Isaan question rides on it.

### RQ2 — Is culinary distinctiveness constituted by inclusion or by exclusion?

Plain: is a region's food about what it uses, or about what it refuses to use?

|                 |                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method          | Decompose each province's distinctiveness into presence-driven and absence-driven components. Validate against interview responses on stated absences.                                                                                                                                                                          |
| Output          | Scatter: presence-driven vs. absence-driven, one point per province, diagonal = balanced. Figure 3.                                                                                                                                                                                                                             |
| A NO looks like | Distinctiveness is symmetric everywhere. Still a fact nobody had.                                                                                                                                                                                                                                                               |
| Transferable    | <b>Every distinctiveness measure in computational humanities — TF-IDF, log-odds, Zeta — is a presence measure. Absence is structurally invisible to all of them.</b> A decomposition that recovers it is a contribution to measurement, and it applies to any corpus where practitioners define themselves against a neighbour. |

This is the most genuinely novel item in the set and it costs almost nothing to compute. The cost is that no off-the-shelf measure exists, so she must define one and defend it — roughly 15 hours of thinking and 3 of coding. Best novelty-to-effort ratio in the project.

### RQ3 — How much of Thailand's culinary map is legible at all from public online data?

Plain: whose cooking did the internet leave out?

|                 |                                                                                                                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method          | Coverage cartography. Labelled fraction, provincial recipe counts, source-domain concentration, provinces clearing each threshold. Run headline analyses at 10 / 15 / 25 recipes and report how conclusions move. Ground-truthed by fieldwork (§8).        |
| Output          | Choropleth with sub-threshold provinces greyed out, plus a threshold-sensitivity panel. Figure 1 — and rhetorically the strongest image in the paper.                                                                                                      |
| A NO looks like | Coverage is uniform and unrelated to periphery. That would validate every scraped-corpus study ever done, which is a real contribution in the opposite direction.                                                                                          |
| Transferable    | <b>Coverage in scraped cultural corpora is structured, not random</b> — it tracks digital access, and therefore compounds existing marginalisation rather than merely adding noise. Papers that do not report it are silently studying the covered subset. |

This converts the project's largest weakness into its most defensible contribution, requires only data she will already have, and states the TunDee thesis in a second domain. **Lead the abstract with this one.**

## RQ4 — Which ingredients hold Thai cuisine together, and which regional cuisines are most fragile?

Plain: what happens if all the garlic in Thailand disappears?

|                 |                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method          | Node-removal robustness on the PMI-weighted co-occurrence network. Delete each ingredient, measure fragmentation of regional subgraphs, rank by structural indispensability. |
| Output          | Horizontal bar chart, faceted by region. Also the site's main interactive.                                                                                                   |
| A NO looks like | All regions equally redundant — no ingredient is structurally indispensable anywhere.                                                                                        |
| Transferable    | Cuisines differ in structural redundancy, with an obvious reading in terms of supply shock and climate risk to a single crop.                                                |

**Status: paper-optional.** Runs in minutes, needs no new data, produces the best interactive on the site. But it is the most decorative of the five and the first to cut if the paper runs long — as a half-page subsection or as future work. Four questions executed cleanly reads better than five with one thin.

## RQ5 — Does regional signal concentrate in vernacular practice rather than in canonical, externally-facing forms?

Plain: is the food Thailand is famous for the least regional food it has?

|                 |                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method          | Classifier run separately per dish category — nam prik, everyday/preserved, curries, restaurant-facing, desserts. <b>Report the majority-class baseline</b> or the accuracy numbers mean nothing.                                           |
| Output          | Grouped bars, accuracy by category, baseline as a dashed line.                                                                                                                                                                              |
| A NO looks like | Signal is uniform across categories — regional identity is distributed evenly through the cuisine rather than concentrated in domestic food.                                                                                                |
| Transferable    | <b>Canonisation flattens variation.</b> The items a tradition exports are the least diagnostic of it — testable in literature, music, and craft, and it predicts that any corpus built from canonical works understates regional diversity. |

**New schema requirement:** a dish-category field on all ~2,200 recipes, with a documented taxonomy of five or six categories and written inclusion rules. Partly automatable from dish names, needs manual review. **Add the field before the March freeze** — retrofitting it afterwards is miserable.

**Read together, the five make an argument rather than running a battery of tests.** Thai regional identity is real and has locatable edges (RQ1); it lives in refusal as much as in addition (RQ2); the internet has flattened it unevenly and worst at the periphery (RQ3); scarcity and structure reveal which parts are load-bearing (RQ4); and the flattening hits hardest exactly where identity is strongest — everyday domestic food (RQ5).

**All five run on the web corpus.** None can be blocked by a cancelled trip, a sparse interview round, or a Grade 12 collision.

## 6. Feasibility Audit — Can These Actually Be Finished? NEW SECTION

An honest question-by-question assessment against the hour budget and the March 2027 freeze. This section exists because v2 had no equivalent, and the temptation with a good question list is to keep all of it.

| RQ  | Verdict                     | Real cost                       | Constraint and mitigation                                                                                                                                                                                                            |
|-----|-----------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ1 | <b>Yes, reduced</b>         | ~25 h analysis                  | Full version testing three boundary sets is not feasible. One comparison only (linguistic). If the labelled fraction is under 35%, this collapses to region level (4–6 units) and the paper's shape changes — knowable now, see §21. |
| RQ2 | <b>Yes, comfortably</b>     | ~18 h                           | Web-only, cheap compute. Cost is definitional, not computational. No blocking dependency.                                                                                                                                            |
| RQ3 | <b>Yes</b>                  | ~20 h + fieldwork               | Built from artifacts she will already have. Fieldwork ground-truthing is additive; without it the coverage result still stands, just without the demonstration of what the blank contains.                                           |
| RQ4 | <b>Yes, trivially</b>       | ~8 h                            | Runs in minutes on the existing network. No new data. Paper inclusion optional.                                                                                                                                                      |
| RQ5 | <b>Yes, at region level</b> | ~25 h labelling + 12 h analysis | Province-level accuracy inside a single dish category will be too thin to report — the data splits five ways. Report at region level. Category labelling is a new, unbudgeted cost.                                                  |

### WHAT WAS CUT, AND WHY

Two questions from earlier drafts do not appear above.

- **Per-province web-vs-fieldwork divergence ranking.** 12–15 interviews across 2 provinces cannot rank 77 provinces by divergence, or even 6. The surviving comparison is **pooled** — web corpus vs. fieldwork corpus overall — with province as context, never as the unit of analysis. Promising a ranking in an abstract would be over-claiming and a reviewer would catch it.
- **Substitution-graph structural equivalence.** Genuinely the most interesting analysis available, and not feasible: 12–15 interviews yield perhaps 40–70 substitution pairs, far too sparse for stable structural equivalence. **Keep collecting the data** — interview question 3 stays in the protocol — describe it qualitatively in the discussion, and put the graph analysis in future work. That is honest and costs nothing.

**The real bottleneck is not analysis.** All five questions together are roughly 110 hours of analysis. Ingredient normalisation alone is ~130 hours and **cannot be delegated** — it is several hundred judgment calls that must be visibly hers. Add ~25 h for dish categories and ~85 h for fieldwork logistics.

Claude Code compresses the machinery and does not compress the judgment. The ~225 hours across 20 human decision gates (§13) is the number that determines whether this finishes.

## 7. Data Stream 1 — Web Scraping SCHEMA ADDITIONS

Sources, yields, and the six-source strategy are unchanged from v2 §5: Wongnai, Cookpad TH, Krua.co and magazine archives, provincial tourism and cultural office pages, regional cookbook OCR, YouTube channel descriptions. Target ~2,200 usable after cleaning, never fewer than four live sources.

The scraping ethics section from v2 is retained **in full and without softening**: robots.txt and ToS checked and dated in [ETHICS.md](#) for every site; rate-limit 1 request / 1–2 sec; honest User-Agent with contact email; ingredient lists and province labels stored, never full recipe prose; raw HTML cached locally and never published.

Two fields to add now — both free today, both unrecoverable later.

- **Publication / post date** on every scraped recipe. Enables any temporal question later ("which ingredients are rising in Thai online cooking") and costs one selector per scraper. Impossible to backfill after the freeze.
- **Dish category**, from the documented taxonomy. Required by RQ5, absent from v2's schema.

**Raw HTML is not stored in the database.** Raw pages are already persisted to disk; duplicating them into Postgres inflates the database for no benefit. The database holds derived structured data only.

### CLEANING PIPELINE — STILL ~40% OF TOTAL HOURS

Unchanged from v2 §5: tokenise with PyThaiNLP [newmm](#) ; normalise to controlled vocabulary; strip quantities and preparation verbs; assign province labels from source page, dish name, or explicit regional claim (**leave unlabelled rather than guessing, and report the unlabelled fraction**); deduplicate on ingredient-set Jaccard above 0.9; build the sparse ingredient × recipe matrix.

The controlled vocabulary remains the project's most defensible contribution: 400–700 normalised Thai ingredient entries with variant mappings, a resource that does not exist publicly and cannot be scraped into existence. Every mapping decision required someone who cooks and reads Thai.

## 8. Data Stream 2 — Fieldwork, Rescoped REWRITTEN

**The role has changed, and so has the size.** In v2 the fieldwork was a held-out test set that two research questions depended on. In v3 no research question depends on it. Its job is now:

1. **Ground-truthing the coverage gap (RQ3).** The coverage map says these provinces are invisible to computational methods. The interviews say and here is what people there actually cook, none of which appears in the corpus. That is an existence claim — exactly what n=12 supports, and unobjectionable because it is not asked to generalise.
2. **Validating the RQ2 measure** against stated absences ("บ้านเราไม่ใส่").
3. **Authorship evidence and the essay.** Unchanged and still irreplaceable.

### REVISED SCOPE

| Parameter     | v2                     | v3                                | Reason                                                                                                                                                                                                    |
|---------------|------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interviews    | 15–25                  | <b>12–15</b>                      | No RQ depends on count. Depth beats spread: one 90-minute interview with photos, a cooked dish and a follow-up beats four rushed 40-minute conversations                                                  |
| Provinces     | 4–6, one per region    | <b>2, deep</b> — Nan and Surin    | 4 interviews per province supports nothing. <b>6 per province supports a specific, defensible sentence:</b> "six of six cooks in Surin named an ingredient the corpus never associates with the province" |
| Trips         | 3–4 weekends + a break | <b>2 trips</b> , one per province | Halves the logistics that compete with Grade 12 coursework                                                                                                                                                |
| Per interview | 40–70 min              | <b>60–90 min</b>                  | The depth the smaller n buys                                                                                                                                                                              |

**Why Nan and Surin.** One Northern, one lower-Isaan; both peripheral; both far from the web corpus's centre of gravity; and Surin sits on the upper-vs-lower Isaan question that RQ1's linguistic comparison turns on. If only one trip happens, **do Surin** — six interviews beats zero by an enormous margin, and Surin carries more analytical weight.

### THE EIGHT-QUESTION PROTOCOL — RETAINED, WITH ONE ADDITION

All eight questions from v2 §6 are kept verbatim, including Q3 (substitutions — still collected even though the graph analysis moved to future work) and Q8 ("what do restaurants get wrong about it?", still the most valuable question in the protocol).

Add question 9, phrased to be asked out loud as written:

“อะไรที่คนจังหวัดอื่นใส่ในงานนี้ แต่บ้านเราไม่มีวันใส่?”

What do people in other provinces put in this that we would never put in?

Q4 captures self-reported boundary markers; Q9 captures **stated absences**, which is what RQ2 validates against. Inferred absence is weak evidence; stated absence is strong. Test the phrasing on the first informant — the wording determines whether the answer is real or merely polite.

**Ethics and consent are unchanged and non-negotiable:** written one-page Thai consent form in plain language; default to anonymised publication by role and province; a parent chaperones every interview and this is stated plainly in the methods section; reciprocity — every participant receives the finished map and a printed copy of their own recipe as it appears in the dataset; no recording without explicit permission.

#### ADDITIONAL FIELDS TO CAPTURE

- **District, not just province.** Isaan provinces are large and sub-provincial resolution costs nothing to record.
- **Ingredient acquisition** — grown / foraged / market / packaged. Enables the wild-ingredient and industrial-penetration observations without a separate study.

## 9. Data Stream 3 — Cook-Alongs, Cut to Eight REDUCED

v2 specified 15 dishes at ~25 hours. v3 specifies **8**, buying back roughly 14 hours for normalisation work, which is the actual bottleneck. Eight is enough to make the origin story evidenced rather than asserted, which is the only job this stream has to do.

Protocol otherwise unchanged: cook from the cleaned dataset's ingredient list rather than the original web page, since the point is to test the pipeline; include at least two dishes the classifier gets wrong; log what was missing, what was substituted, whether the result matched expectation, and one sentence of sensory description; photograph the mise en place rather than only the finished dish; where a dish came from an interview, cook that version and send the photograph back to the cook.

The scope-discipline warning from v2 stands and matters more with a smaller number: this is not a cookbook, a food blog, or a YouTube channel. The cooking serves the data. The moment it becomes the deliverable, the project has drifted into a food blog with a statistics section.

## 10. Figure Specification — Seven Figures NEW SECTION

Specified before implementation, with axes named. v2 had no figure specification, which is how projects end up with a UMAP plot doing work it cannot do.

| # | Figure                               | X axis                             | Y axis                                                      | Encoding and purpose                                                                                                                                                                                                                                         |
|---|--------------------------------------|------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Coverage map (RQ3)                   |                                    | geography                                                   | Fill = recipe count; sub-threshold provinces greyed. Paired with a sensitivity line chart: X = threshold 5–30, Y = provinces qualifying. <b>An image of Thailand with a third of it blank is the paper's most effective figure.</b>                          |
| 2 | Distance-decay (RQ1)                 | km between province pair (0–1,800) | cosine distance on TF-IDF (0–1)                             | ~2,900 points, one per pair. Colour = same-region vs. cross-region. LOESS fit with change-point marked. Outliers below the line are far-apart provinces that cook alike — each one is a story.                                                               |
| 3 | Distinctiveness decomposition (RQ2)  | presence-driven distinctiveness    | absence-driven distinctiveness                              | One point per province; diagonal = balanced. Above the line = defines itself by refusal. A chart type reviewers have not seen before.                                                                                                                        |
| 4 | Prevalence vs. distance (supporting) | km from Bangkok                    | share of province's recipes containing the ingredient (0–1) | Small multiples, 6–12 panels, shared axes, colour = region. Include one flat panel (MSG) as a null control so readers can calibrate what "no pattern" looks like. <b>Most explanatory figure in the set; first to move to an appendix if space is tight.</b> |
| 5 | Fragility ranking (RQ4)              | fragmentation caused on removal    | ingredient, sorted                                          | Horizontal bars faceted by region. Also the site's main interactive.                                                                                                                                                                                         |
| 6 | Category accuracy (RQ5)              | dish category                      | classification accuracy                                     | Grouped bars with the majority-class baseline as a dashed horizontal line. Five bars, one line, immediately readable.                                                                                                                                        |
| 7 | Province × ingredient heatmap        | ingredients (top ~60 by variance)  | provinces                                                   | Fill = TF-IDF. <b>Both axes seriated by the hierarchical clustering, never alphabetical.</b> Blocky diagonal structure = regional cuisines exist. Good first figure in Results because no modelling sits between data and image.                             |

**The ordering trap.** The confusion matrix and the heatmap must both be ordered **geographically or by cluster, never alphabetically**. Alphabetical ordering scatters real structure into what looks like noise. This single choice is the difference between a figure that is the result and a figure that says nothing.

**A prototype of Figure 4 already exists** as an HTML mockup built on synthetic data — structure only, no findings. It is a structural prototype for the real build and should be regenerated from the ~300 SorKorPor recipes as the first signal check (§21).

## 11. Technical Architecture REVISED

Five stages, each a separate notebook, each producing a checkpointed artifact the next consumes — unchanged in principle from v2 §10. What changed is the storage layer and the serving layer.

| Component          | Choice                                                                       | Why                                                                                                                                                                                                                                             |
|--------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database           | <b>Local PostgreSQL 15 + PostGIS, in Docker Desktop</b> (Mac, Apple Silicon) | Replaces the earlier hosted plan, which hit a storage cap and added latency to every iteration. Local is faster, free, and has no quota. PostGIS handles province geometry, centroids, and contiguity weights natively                          |
| Raw pages          | On disk, gitignored, <b>not in the database</b>                              | Already persisted to disk; duplicating into Postgres inflates the DB for no benefit                                                                                                                                                             |
| Production API     | <b>Pre-materialised static JSON</b> , not live Postgres queries              | The public site's data never changes after the freeze. Static JSON on a CDN is faster, free, and cannot go down because a container stopped. Removes the site's only runtime dependency                                                         |
| Language / env     | Python 3.11, local venv, pinned <code>requirements.txt</code>                | Reproducibility — a reviewer must be able to re-run it                                                                                                                                                                                          |
| Heavy compute      | None required                                                                | UMAP on $2,200 \times 700$ is trivial and nothing needs a GPU. <b>Say so in the paper</b> — it establishes that the project was constrained by effort and access rather than resources, which is the profile of a student project done properly |
| Thai NLP / spatial | PyThaiNLP; PySAL, esda, geopandas                                            | Unchanged                                                                                                                                                                                                                                       |
| Visualisation      | matplotlib + seaborn for the paper; D3 or Leaflet for the site               | Separate publication figures from the public artifact — different audiences                                                                                                                                                                     |
| Site hosting       | GitHub Pages or Vercel free tier                                             | Zero cost, zero maintenance                                                                                                                                                                                                                     |

Total infrastructure cost remains effectively zero. Real costs are fieldwork travel (now roughly 5,000–9,000 THB across two trips, down from v2's four) and cook-along ingredients.

The repository structure from v2 §9 is retained, with three additions: `docs/decisions.md`, `docs/hypotheses.md`, and `data/coverage/`.

## 12. Analysis Plan CORRECTIONS APPLIED

### 12.1 THE INGREDIENT NETWORK

Unchanged from v2 §11.1. Nodes are normalised ingredients; edges join ingredients appearing in the same recipe; **weight by pointwise mutual information rather than raw co-occurrence**, or the graph is dominated by fish sauce, garlic and chilli and tells you nothing. This choice is worth a paragraph in the paper. Extract the backbone by disparity filtering before visualising. Run Louvain at several resolution parameters and report community stability across them. Compare to conventional regions with adjusted mutual information. Name communities from their highest-PMI ingredients and check the names against what interviewees said.

### 12.2 GEOGRAPHY — WITH THE CORRECTIONS FROM §4

| Analysis         | Specification                                                                            | Trap to avoid                                                                                                                                                                                       |
|------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Province vectors | TF-IDF over ingredients per province. Threshold swept, not chosen (RQ3)                  | Do not present a 77-province map if only 40 cleared the threshold. Grey the rest and say why                                                                                                        |
| UMAP             | Project province vectors to 2D. Colour by region                                         | <b>Axes carry no interpretable meaning.</b> Never quantify from the projection, never label an axis geographic or cultural, never run Moran's I on a UMAP dimension. Browsing interface, not result |
| K-means          | Sweep k from 2 to 10; report silhouette scores                                           | Choosing k=4 because Thailand has four regions is assuming the answer                                                                                                                               |
| Moran's I        | <b>Interpretable scalars only.</b> Queen-contiguity and k-nearest weights, both reported | Sensitive to the weights matrix; run both or a reviewer will ask. Supporting analysis inside RQ1, not a headline                                                                                    |
| Mantel           | Cosine culinary distance vs. great-circle distance. <b>9,999 permutations</b>            | <b>Never a parametric p-value.</b> Matrix entries are not independent. A significant result is also not causal — distance may proxy for shared history, climate, or trade                           |

### 12.3 THE CLASSIFIER AND ITS CONFUSION MATRIX

The accuracy number is the least interesting output; the confusion matrix is the finding. Group to region level as well as province level — province-level accuracy will be low and that is expected, not a failure. **Report the majority-class baseline.** Beating it is the real claim. Order the matrix geographically. Extract feature importances and check the top discriminating ingredients against what interviewees named: agreement is triangulation, disagreement is a discussion section.

13. Human Decision Gates & the Claude Code Arrangement

NEW SECTION

**The arrangement, stated plainly.** Claude Code builds machinery. She owns every analytical decision. Twenty human decision gates (HD-1 ... HD-20) mark the points where the pipeline stops and waits for a judgment only she can make — and every one of those judgments is written up in `docs/decisions.md` with its reasoning and its alternatives.

That file is not bookkeeping. It is **interview preparation, methods-section source material, and the single best answer to "how much help did you have?"** A student who can open a dated file and show two hundred documented judgment calls is not answering that question defensively.

Representative gates: the canonical-vocabulary granularity decision (are three chillies one ingredient or three?); the province-labelling rules; the minimum-recipe threshold; the dish-category taxonomy and inclusion rules; the definition of the RQ2 distinctiveness decomposition; the choice of competing boundary set for RQ1; which dishes to cook; which provinces to visit; what goes in the limitations register.

**~225 hours across the twenty gates.** This is the number that determines whether the project finishes, and it is the one number that generating code faster does not reduce. Plan the calendar around it, not around compute.

**AI disclosure is a credibility asset, not a risk.** v2 §16 is right and v3 strengthens it: a line reading "I used Claude Code to build the pipeline and to debug the tokeniser; every analytical decision is mine and documented with its reasoning in docs/decisions.md" is stronger than silence. In 2027 every reader assumes AI assistance was available; the discriminating question is whether the student is specific about it.

14. The Limitations Register

EXTENDED

Maintained as a live file from month one, not written at the end. Every v2 entry is retained: web corpus skews urban and Central Thai (HIGH); province labels sparse and partly inferred (HIGH); uneven coverage across 77 provinces (HIGH); small fieldwork n (MEDIUM); normalisation involves judgement calls (MEDIUM); recipes are not consumption (MEDIUM); modern recipes flatten historical variation (MEDIUM); sites excluded for ToS reasons (LOW).

CHANGES AND ADDITIONS

| Entry                  | Severity                     | Statement                                                                                                                                                                                                             |
|------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single annotator       | LOW-MED →<br><b>RESOLVED</b> | No longer a limitation. 100 recipes independently double-labelled; Cohen's $\kappa$ reported. Cheapest credibility in the project — a few hours of a second Thai reader's time                                        |
| Fieldwork n = 12–15    | MEDIUM                       | <b>Reframed.</b> Presented as an existence demonstration and qualitative validation, never as a representative sample and never as a basis for ranking provinces. Two provinces at $n \approx 6$ each, stated as such |
| Dish-category taxonomy | MEDIUM<br><b>NEW</b>         | Categories are constructed, not natural. Publish the taxonomy and inclusion rules so others can disagree with specific assignments                                                                                    |

| Entry                      | Severity      | Statement                                                                                                                                                                                                                   |
|----------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No temporal dimension      | MEDIUM<br>NEW | <b>The corpus is a single time-slice.</b> She can honestly say traditions are undocumented; she cannot say how fast they are vanishing. Any claim about rate of loss is unsupported — catch this before it reaches an essay |
| Linguistic distance matrix | MEDIUM<br>NEW | Constructed from published ethnolinguistic maps, with judgment calls at every boundary. Publish the matrix                                                                                                                  |

## 15. The Paper — Venue and Structure RETARGETED

**Working title:** "What 2,200 online recipes do and do not reveal about regional boundaries in Thai cooking" — retained from v2. Accurate, interesting, and immune to the obvious objection.

| Venue                                  | Track               | Assessment                                                                                                                                                                                                                                                                         |
|----------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journal of Cultural Analytics          | Primary             | <b>v2 said do not attempt. That is now wrong.</b> With a March 2027 freeze and November 2027 applications, a review cycle fits. "Under review at a peer-reviewed journal" is a stronger honours line than a workshop poster, and JCA is the natural home for the RQ3 methods claim |
| CHR 2028                               | Full paper / poster | Computational Humanities Research is the right community for this work. Timing suits a post-journal submission or a parallel short paper                                                                                                                                           |
| arXiv (cs.CY / cs.SI)                  | Preprint            | <b>Post regardless of venue outcome, before applications.</b> A citable preprint with a DOI is what the Common App honours line actually needs                                                                                                                                     |
| Thai undergraduate research conference | Full paper          | Realistic, genuine, in Thai or English. Local recognition strengthens the EEF and Chulalongkorn relationships                                                                                                                                                                      |

### STRUCTURE — ~4,600 WORDS

| Section                 | Words | Notes                                                                                                                                                                                                                                                               |
|-------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Introduction            | 600   | Opens with the kitchen. The origin story is the introduction — do not replace it with a literature review                                                                                                                                                           |
| Related work            | 500   | Ahn et al. 2011 flavour networks; computational gastronomy; Thai food studies; DH regional-variation work. <b>State the disagreement:</b> co-occurrence networks are structurally blind to substitution, and standard distinctiveness measures are blind to absence |
| Data                    | 900   | Sources, scraping ethics, cleaning pipeline, the lexicon, the fieldwork protocol. Longest methods subsection, because the data is the contribution                                                                                                                  |
| Methods                 | 700   | Justify PMI weighting, the k sweep, the permutation Mantel, and the threshold sweep explicitly                                                                                                                                                                      |
| Results                 | 1,100 | Negative results at full weight, not in a footnote. Compare against the pre-registered expectations                                                                                                                                                                 |
| Ground-truthing the gap | 400   | <b>The distinctive section.</b> Two provinces, twelve cooks, and a table of dishes and ingredients absent from the corpus. No other computational gastronomy paper has this because no other author did the interviews                                              |
| Limitations             | 500   | Lifted from the live register. Unusually long, deliberately                                                                                                                                                                                                         |
| Conclusion              | 300   | What she would do next and what she would do differently                                                                                                                                                                                                            |

**Authorship rules are unchanged from v2 and non-negotiable:** she is first author, and if an arrangement would not permit that, decline the arrangement. A faculty advisor is a co-author only for substantive contribution to design or analysis — review and advice is an acknowledgement. Getting that boundary right is itself a mark of research maturity. The sister is acknowledged by name for visualisation and design.

## 16. OpenFlavorTH — Two Datasets, Not One SPLIT

**Release the lexicon separately.** The recipe corpus is competent but modest — comparable public food datasets run to hundreds of thousands of recipes, and a realistic first-year expectation here is **150–600 downloads**. The lexicon is a different proposition: it is useful to Thai NLP, recipe parsing, food-safety text mining, and agricultural translation — people who do not care about regional cuisine at all.

**Two repos:** `OpenFlavorTH-recipes` and `OpenFlavorTH-lexicon`. Bundling them hides the lexicon's independent value and prevents it being cited on its own.

| Artifact                                   | Contents                                                                                                                                                                                                              |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>recipes.parquet</code>               | Recipe ID, normalised ingredient list, province label (or null), region label, <b>dish category</b> , source domain, <b>publication date</b> , collection date. No recipe prose, no instructions, no copyrighted text |
| <code>ingredients_lexicon.csv</code>       | Canonical Thai name, English gloss, all observed variants, category, notes on judgement calls. <b>The flagship artifact</b>                                                                                           |
| <code>coverage.csv</code> <span>NEW</span> | Recipe count, labelled fraction, and source-domain concentration per province. First-class file, because RQ3 is built on it                                                                                           |
| <code>network.gexf</code> <span>NEW</span> | PMI-weighted co-occurrence network. Input to the RQ4 fragility analysis and the thing a reuser most wants                                                                                                             |
| <code>province_vectors.csv</code>          | TF-IDF flavour vector per qualifying province, with recipe counts                                                                                                                                                     |
| <code>fieldwork.csv</code>                 | Anonymised interview recipes: province, district, role of cook, ingredients, stated substitutions, stated absences, acquisition mode                                                                                  |
| Dataset cards                              | Motivation, collection method, scraping ethics, known biases, unlabelled fraction, κ, licence, citation, contact                                                                                                      |

**Because all five research questions are answered from the released artifacts, the dataset is the paper's reproducibility layer**, not a side deliverable. A reviewer can re-run every analysis from the public files. That is a stronger position than most published work in the field, and it should be said in the dataset card.

### RELEASE CHECKLIST

- CC-BY-4.0 for data, MIT for code, stated in three places.
- Zenodo DOI via the GitHub link — what makes the Common App honours entry verifiable.
- A `loading_script` so `load_dataset` works in one line. Small effort, large effect on adoption.
- **English glosses on every lexicon entry** — opens it to non-Thai readers, who are most of the potential users.
- Croissant metadata file.

- One honest post to PyThaiNLP community channels at launch. Not a campaign.
- Version it: v1.0 at submission, v1.1 if a second fieldwork round lands.
- **Report the real download number at application time.** "340 downloads" is far more credible than "widely used". If it is small, let the lexicon's usefulness carry the claim.

## 17. The Public Artifact LARGELY UNCHANGED

Thai-first, phone-first, shared through Line — the reasoning in v2 §15 stands. What gets built: the map (tap a province, see distinctive ingredients, nearest culinary neighbours which are often not geographic neighbours, and one dish; greyed-out provinces explicitly labelled "not enough data", because the honesty is part of the design); the province quiz with a shareable result card; the build log linked from the footer; and a plain-language methods page linked from every map view.

**One addition:** the RQ4 fragility interactive — pull an ingredient out of the network and watch it fragment. It is the best demo in the project and it is the thing people forward to each other.

**Ship the map, the quiz, and one interactive. Nothing more.** Site builds reliably take triple the estimate, and this one competes directly with the analysis window.

Deliberately not building, unchanged from v2: a recipe app or recommendation engine; a native mobile app; user accounts or saved profiles (collecting personal data from Thai minors creates real obligations for no research benefit); a restaurant or ingredient-sourcing directory.

## 18. The Credit Ledger ONE ROW UPDATED

v2 §16 is retained in full — the principle that unnamed help looks like hidden help, and that naming exactly who did what converts suspicion into confidence. Published as [CREDITS.md](#) and reproduced in the paper's acknowledgements. Her: all scraping, cleaning, lexicon construction, analysis, modelling, writing, fieldwork, transcription, cooking, and every build-log post. Advisor: named, bounded, hours stated. Sister: named, for visual identity, result-card design, map interface, Line distribution, photography. Parents: chaperoned, transported, made two introductions — did not conduct interviews, analyse data, or write. Participants: by role and province, or by name where consent was given.

**The AI row is updated** to reflect the Claude Code arrangement, and should read approximately: Claude Code was used to build the data pipeline, the scrapers, and the analysis machinery, and to debug the Thai tokeniser. Every analytical decision — vocabulary granularity, labelling rules, thresholds, taxonomy, model choices — is mine and is documented with its reasoning in docs/decisions.md. Claude Code did not choose what to measure, did not interpret any result, and did not write the paper.

## 19. Timeline to November 2027 REWRITTEN

Anchored on three fixed dates: **dataset freeze 31 March 2027**, **applications November 2027**, and the ISB school calendar — which puts intensive work in the June–August break rather than in term time. Grade 12 begins August 2027, so the writing must be substantially done before it starts.

| Window       | Build                                                                                                                                                                | Fieldwork / cooking                                                                        | Output                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Aug–Sep 2026 | <b>Three blocking items (§21).</b><br>Labelled-fraction measurement. Chart 4 signal check on 300 SorKorPor recipes. Postgres/PostGIS container running. Scrapers 1–3 | Cook the origin dish with family. Book both trips into the family calendar with real dates | Repo public. <a href="#">docs/hypotheses.md</a> committed. Build-log post 1. <b>Go / no-go on province-level analysis</b> |

| Window                                   | Build                                                                                           | Fieldwork / cooking                                      | Output                                                                                                    |
|------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Oct-Dec 2026                             | Corpus to ~1,400. Tokenisation and normalisation. Lexicon v0.5. Dish-category taxonomy defined  | Trip 1 — Surin, 6 interviews. Cook 2 dishes              | Post 2. Lexicon first release. <a href="#">ETHICS.md</a> complete                                         |
| Jan-Mar 2027                             | Corpus to ~2,200. Normalisation complete. Category labelling. Second-annotator κ on 100 recipes | Trip 2 — Nan, 6 interviews. Transcription. Cook 3 dishes | Post 3 (failure post). <b>FREEZE 31 MARCH</b>                                                             |
| Apr-May 2027                             | RQ3 coverage cartography. RQ1 distance-decay + change points. Network, Louvain, backbone        | Cook 3 dishes, including two the classifier gets wrong   | Post 4. Figures 1, 2, 7                                                                                   |
| Jun-Aug 2027<br>school break — intensive | RQ2 decomposition. RQ4 fragility. RQ5 classifier + baseline. Full analysis complete             | Send results back to participants                        | Post 5. Figures 3–6. <b>Paper drafted. Dataset packaged</b>                                               |
| Sep-Oct 2027<br>Grade 12 begins          | Advisor review. Revisions. Site and quiz built with sister                                      | —                                                        | Post 6 (limitations). <b>Journal submission. arXiv preprint + Zenodo DOI. Both HuggingFace repos live</b> |
| Nov 2027                                 | Line launch, timed to a Thai food moment                                                        | —                                                        | <b>Applications. Materials complete</b>                                                                   |

**Deliberate slack** sits in Oct-Dec 2026 and Sep-Oct 2027, both of which collide with coursework. School always wins: a 5 in Calculus BC matters more to every target programme than a month of this project.

## 20. Risk Register REPRIORITISED

| Risk                                                | Severity             | Mitigation                                                                                                                                                                                                                                                           |
|-----------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Province labels too sparse for any geographic claim | HIGH                 | <b>Now the top risk, replacing fieldwork.</b><br>Measure the labelled fraction on the 300 SorKorPor recipes this month. If below 35%, pivot to region level (4–6 units) and say so openly. Everything downstream branches on this number and it is currently unknown |
| Normalisation grind stalls the project              | HIGH                 | ~130 hours that cannot be delegated. Will feel endless around month 4. Fix a weekly quota, work in 50-ingredient batches, and log every decision to <code>docs/decisions.md</code> so the hours visibly produce an artifact rather than disappearing                 |
| Origin-story bracket never filled                   | HIGH                 | Two hours of work that gates the essay, the introduction, and every interview answer. Cannot be reconstructed later. See §3 and §21                                                                                                                                  |
| Fieldwork does not happen                           | MEDIUM<br>was HIGH   | <b>Downgraded by design.</b> No RQ depends on it. Paper loses the ground-truthing section and the authorship story weakens, but stands. If only one trip is possible, do Surin                                                                                       |
| Dish-category labelling underestimated              | MEDIUM<br><b>NEW</b> | ~25 unbudgeted hours, and RQ5 dies without it. Automate from dish names first, review manually second. Must be inside the March freeze                                                                                                                               |
| Grade 12 workload collision                         | MEDIUM               | Timeline puts analysis in the summer break and leaves Sep–Oct light. School wins                                                                                                                                                                                     |
| Results are null                                    | MEDIUM               | Handled by design: all five questions are framed so a null answer is reportable, and the pre-registration makes the negative result read as a finding                                                                                                                |
| Scope creep into cooking or site-building           | MEDIUM               | Cook-alongs capped at 8. Site is a map, a quiz, and one interactive. Re-read §9 and §17 when either starts feeling like the deliverable                                                                                                                              |
| A key site blocks scraping                          | MEDIUM               | Six sources, never fewer than four. Cache raw HTML on first pass                                                                                                                                                                                                     |
| Journal rejection                                   | LOW-MED              | arXiv preprint, Zenodo DOI, and two public datasets already satisfy the honours entry. CHR 2028 remains open. Submission is a bonus, not a dependency                                                                                                                |
| Advisor unavailable                                 | LOW                  | Project is completable without one. The advisor improves the paper; they are not a critical path                                                                                                                                                                     |

## 21. The Next 30 Days — Three Blocking Items NEW

**Five rounds of question refinement have happened. Zero measurements.** The questions are well past the point of diminishing returns and the data is unexamined. Nothing else in this document should proceed until these three are done. None takes more than an afternoon.

| # | Action                                                                                                                                                                                           | Cost  | Why it blocks everything else                                                                                                                                                                       |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Measure the labelled-recipe fraction</b> on the ~300 SorKorPor recipes already in hand. One number: what share carry a usable province label?                                                 | 2 hrs | Determines whether this is a province-level paper (>35%), a region-level paper, or primarily a coverage paper. RQ1's entire form depends on it. Currently unknown, and knowable today               |
| 2 | <b>Build Figure 4 from real data</b> — prevalence vs. distance from Bangkok, on the same 300 recipes. A groupby and a scatter                                                                    | 3 hrs | If pla ra and sticky rice separate at n=300, the signal is real and the next fifteen months are worth committing. If every panel looks like the MSG control, better to know in month 1 than month 8 |
| 3 | <b>Cook the origin dish with the family member.</b> Photograph the mise en place. Write down what actually happens, including what contradicts the remembered version. Commit with the real date | 2 hrs | The one item that cannot be reconstructed later. The essay, the paper's introduction, and every interview answer currently rest on an unfilled bracket                                              |

Then, in the same month and in this order: make the repo public if it is not already (the first commit date is a timestamp that can never be created retroactively); commit `docs/hypotheses.md` with predictions for all five RQs, dated, before any analysis; put both fieldwork trips in the family calendar with real dates; write the protocol and the Thai consent form; and start `CREDITS.md`, `LIMITATIONS.md`, and `decisions.md` as living documents that get appended to all year rather than written at the end.

## 22. Application Language & the TunDee Pairing REVISED

### Common App activity entry (150 characters):

"Scraped 2,200 Thai recipes, interviewed 12 cooks in 2 provinces, measured what the internet leaves out. Open dataset + paper." — 121 characters. Same facts as v2, with the RQ3 framing that makes the project one sentence rather than a list.

**Interview answers to prepare** are unchanged from v2 §19, with one update: "how much help did you have?" is now answered by opening `docs/decisions.md`. She should be able to name the mentor, the advisor, the sister, the parents, and the Claude Code arrangement with bounded roles in twenty seconds without hesitation. Hesitation is what costs.

**The connective tissue with TunDee — say this in every interview.**

The data that exists about a population is not the truth about that population,  
and the gap widens with distance from the centre.

TunDee closes that gap in scholarship access. FlavorMap measures it in food. One is intervention, one is measurement, and together they are a coherent research identity rather than two impressive hobbies.

**Put that sentence, verbatim, in both projects' public materials** — the FlavorMap README and the TunDee site. If a reader meets the same claim twice in two domains, the identity demonstrates itself, and she never has to assert in an essay that she has one. Asserted, it reads badly; demonstrated, it reads as inevitable.

**RQ3 sharpens the pairing.** A map of Thailand with a third of the provinces greyed out because nobody put their food on the internet is the same picture as the scholarship-access gap, in a different domain. That is the slide, and it is the answer to "why does this matter?"

**One caution on weighting.** If FlavorMap is the weaker of the two projects, it should be the supporting item — one strong activity entry plus a mention inside the TunDee essay — not a second full-strength narrative competing for the same space. And keep the sister's contribution visible: two named teenage siblings shipping a Thai-language public tool is a materially harder story to disbelieve than one student with two impressive projects.

**The test this document is still built around.** Hand the finished project to someone who has never met her and ask one question: **can you picture a sixteen-year-old doing this on weekends?**

With the repository history, six build-log posts, twelve interviews conducted in Thai in two provinces, eight dishes cooked, a named credit ledger, a pre-registration written before the analysis, two hundred documented decisions, and a limitations register she wrote before anyone asked — the answer is yes. The technical work never changed between v1 and v3. Only the evidence did, and the evidence was the part that was missing.